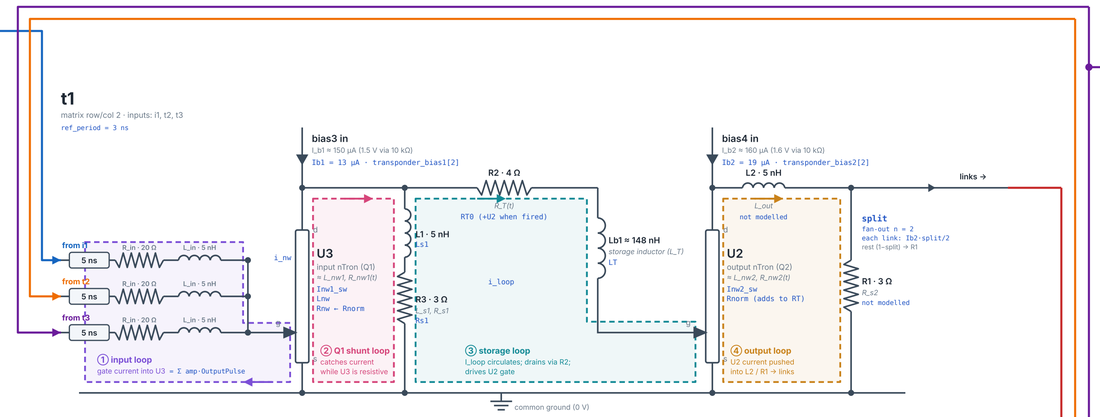
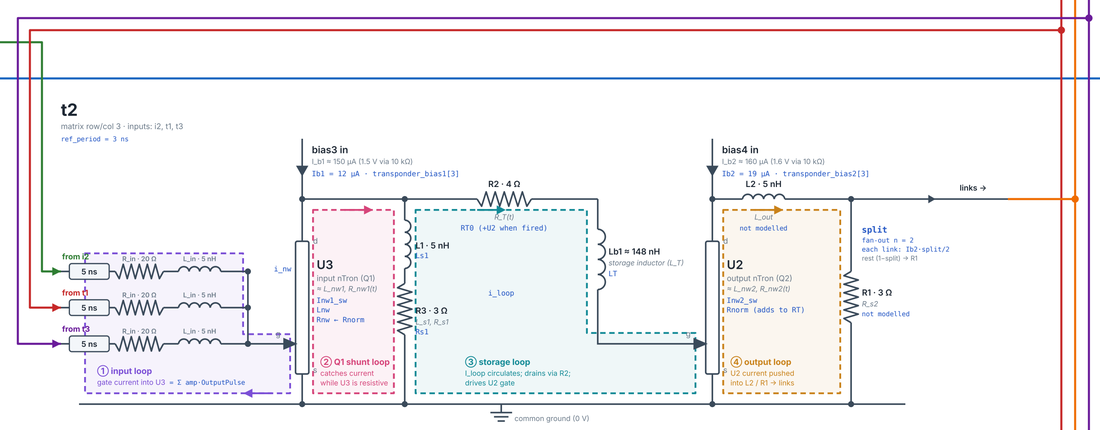
Six transponders · the toy network drawn as six copies of the single-transponder schematic

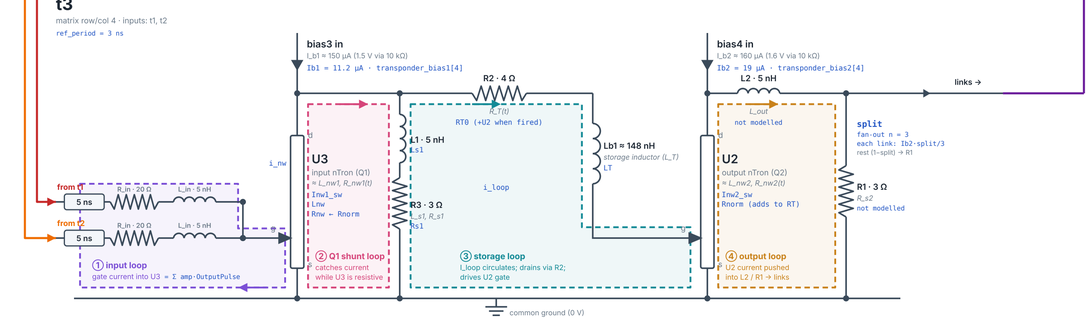
Every box is one copy of single_transponder_simple.pdf; the pulsers are replaced by the links. Wire colour = sending node. Blue monospace = variable names in toy_example.py / notebook N06.

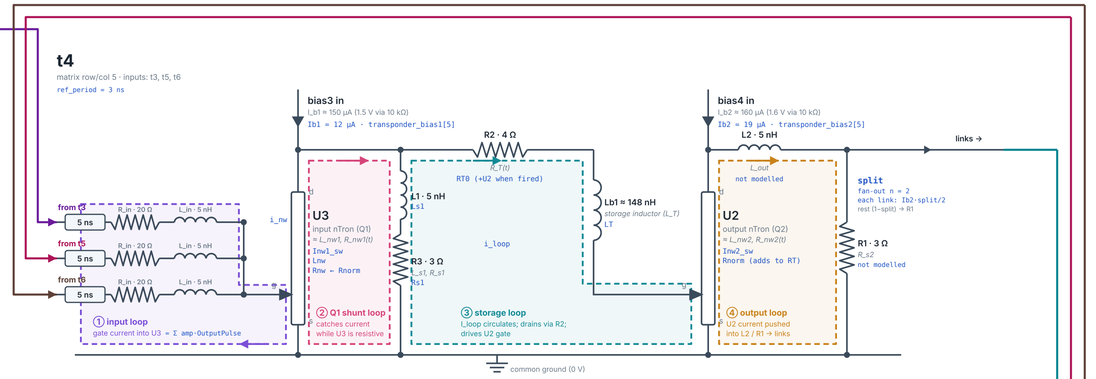
i1 · pulser1 (U1)
sensor click at 1 ns
`n = 2, same Ib2·split rule`

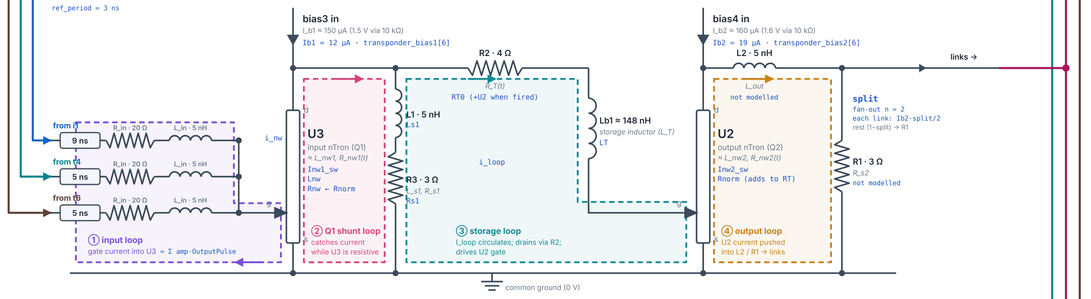
i2 · pulser2 (U4)
click at 1 ns + dt
`n = 2, same Ib2·split rule`

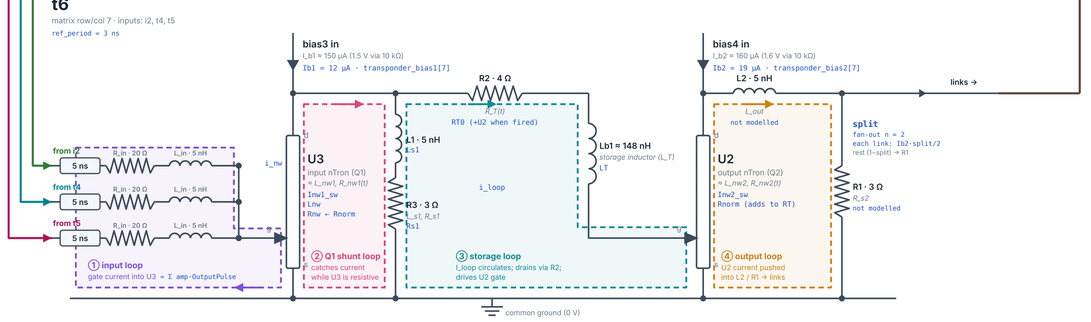












Legend

from i1 · from i2 · from t1 · from t2 · from t3 · from t4 · from t5 · from t6

Loops (same as the single-transponder drawing):

① input loop: gate current into U3, fed by the links (R_in/L_in) ② shunt loop: catches Ib1 while U3 is resistive
③ storage loop: I_loop circulates through Lb1 and R2 and drives the gate of U2 ④ output loop: U2 current pushed into L2/R1 and out on the links

Things to know:

· split enters at one place only: each output junction. Each link gets Ib2·split/n of nominal amplitude (peak ×0.61 after OutputPulse); the rest goes to the R1 shunt.
· The LTspice drawing has 2 inputs (R28/L18, R29/L19). The toy network needs 3 inputs for t1, t2, t4, t5, t6, so the input branch is repeated per link.
· LTspice values: Ib1 ≈ 150 µA, Ib2 ≈ 160 µA, R2 = 4 Ω, R3 = 3 Ω. Toy values: Ib1 = 12–13 µA, Ib2 = 19 µA, RT0 = 3 Ω, Rs1 = 2 Ω (notebook defaults).
· Sensors are drawn as the ideal pulsers of the original drawing, but the toy gives them the same output rule as a transponder (Ib2, split).
· L2, R1, L_nw2 and the R_in/L_in values are not variables in the toy; the delay boxes are the delay matrix entries; all ground rails are the same 0 V node.